Retail Pricing Optimization Analysis

Phase 2: SQL Validation Report

Project Background:

This project analyzes how discounting affects profitability using the Kaggle Superstore retail dataset.

Phase 1 focused on Excel-based exploratory analysis.

The main finding from Phase 1 was:

Profitability becomes negative after the 21–30% discount range.
Phase 2 was completed to validate this finding using SQL and to uncover deeper regional and transaction-level insights using advanced query analysis.

Objective of Phase 2:

The purpose of this phase was to validate Excel findings using SQL and answer the following business questions:

- Does SQL confirm the discount-profit relationship found in Excel?
- Which regions are most sensitive to discounting?
- How many transactions fall into risky discount ranges?
- Are product categories responsible for profitability issues?
- Does profitability vary across time?

SQL Skills Applied:

This phase included practical use of:

- CASE WHEN
- INNER JOIN
- GROUP BY
- Aggregate Functions
- Common Table Expressions (CTE)
- Window Functions
- Ranking Functions

Dataset Structure Used

The original flat dataset was transformed into relational tables.

Customer Table
Contains customer information
Orders Table
Contains:

- Order details
- Region information

- Date information

Product Table

Contains:

- Product information
- Category
- Sub-category

Transactions Table

Contains:

- Sales
- Quantity
- Discount
- Profit

This table was used as the primary analytical table.

SQL Analysis Performed

Analysis 1: Discount Band Validation

Purpose

To validate whether profitability decreases as discount increases.

Finding

SQL analysis confirmed that average profit decreases as discount increases.
Profitability remained positive at lower discount levels.
Profitability became negative after the **21–30% discount range**.
Losses became larger in discount bands above 30%.

Insight

This confirms the main Excel finding.
The business has a clear profitability threshold.
Discounting beyond this level directly harms profit.

Recommendation

The company should maintain a standard discount ceiling of **20%**.
Discounts above this level should require review.

Analysis 2: Category-Level Profitability

Purpose

To determine whether losses are concentrated in specific product categories.

Finding

All categories showed profitability decline at higher discount levels.
However, all categories remained profitable overall.

Total category profit:
Furniture: 285204.72
Office Supplies: 518473.83
Technology: 663778.73

Insight

The issue is not category weakness.
The issue is discounting strategy.
Even strong categories lose profitability when discounts become aggressive.

Recommendation

Discount control policies should be applied across all categories.

Analysis 3: Regional Discount Sensitivity

Purpose

To evaluate regional response to discount increases

.Finding

Most regions became loss-making after the **21–30% discount range**.
However, North Asia became loss-making earlier, at the **11–20% discount range**.

Insight

North Asia is more sensitive to discounting than other regions.
Even moderate discounting reduces profitability in this market.

Recommendation

North Asia should have stricter pricing controls.
Recommended discount ceiling: 10%

Analysis 4: Regional Profitability Ranking

Purpose

To identify highest-risk and lowest-risk regions.

Finding

Regional risk ranking showed:

Lowest profitability:

1. . Southeast Asia
2. EMEA
3. Africa

Highest profitability:
North Asia

Insight

This reveals two different regional patterns.
North Asia performs strongly overall but reacts negatively to discount increases.
Southeast Asia shows weak profitability even without extreme discounting.

Recommendation

The company should not use one universal pricing strategy.Different regions require different discount policies.

Analysis 5: Risky Transaction Analysis

Purpose

To measure the scale of high-risk transactions.

Finding

SQL identified: **11,328 risky transactions**
Average loss per risky transaction: **-71.92**

Insight

This is not a small operational issue.
The business is processing a large number of transactions in loss-making discount ranges.
This indicates a systematic pricing inefficiency.

Recommendation

Implement automated monitoring for high-discount transactions.
Transactions above threshold should be flagged for review.

Analysis 6: Monthly Profitability Trend

Purpose

To check whether profitability changes across time.

Finding

Monthly average profit fluctuated across the year.
Lowest month: **Month 9**
Highest month: **Month 2**

Insight

Profitability is influenced by time-based patterns.
This suggests possible seasonal pricing behavior

Recommendation

Future pricing strategy should consider monthly performance trends.

Major Insights from SQL Phase

Insight 1
SQL fully validated the Excel profitability threshold.

Insight 2
The key profit-loss break point is the **21–30% discount range**

Insight 3
North Asia becomes loss-making earlier than other regions.

Insight 4
Southeast Asia has the weakest overall profitability.

Insight 5
11,328 risky transactions are reducing business profitability.

Insight 6
The problem is pricing strategy, not weak product categories.

Final Business Recommendations

1. Company-Wide Discount Limit
Keep standard discounts below **20%**

2. North Asia Pricing Rule
Limit discounts to **10%**

3. Region-Specific Pricing Strategy
Avoid applying identical discount policies across all regions

4. High-Discount Monitoring
Review all transactions above the profitability threshold

5. Monthly Profit Tracking
Adjust pricing strategy based on monthly trends

Final Conclusion

Phase 2 successfully validated all major findings from Excel analysis.
SQL analysis confirmed that aggressive discounting is the primary cause of profit loss.

It also revealed regional differences in discount sensitivity and identified large-scale risky transactions.

The findings support a smarter, region-specific pricing strategy.

Next Phase

Phase 3 will focus on Power BI dashboard development.

The goal of this phase will be to convert analytical findings from Excel and SQL into an interactive business intelligence dashboard for decision-making.

Planned dashboard development includes:

- Executive KPI Cards
- Profit by Discount Band Visualization
- Regional Profitability Dashboard
- Category Performance Analysis
- Monthly Profit Trend Dashboard
- High-Risk Transaction Monitoring
- Interactive Filters and Slicers

The dashboard will help transform raw analytical findings into business-friendly visual insights for pricing strategy decisions.